

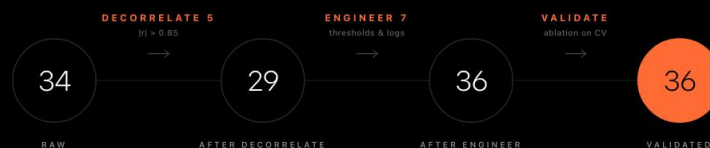
WIDS GLOBALTHON · WILDFIRE RISK

Better features.

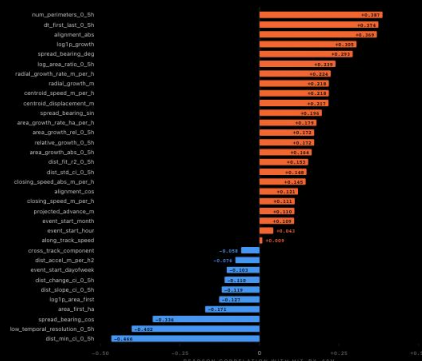
Better model.

THE RECIPE

Screen. Decorrelate. Engineer. Validate.



All 34 features by Pearson correlation with `hit_by_48h`. **Orange** = positive (more likely to hit) · **Blue** = negative (less likely). A linear screen — low correlation isn't a verdict, since trees can still find non-linear signal.



Some signals are the *same* signal.

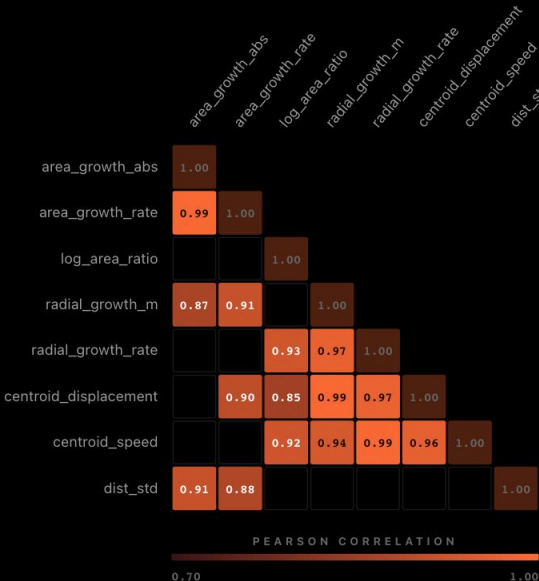
Across the 34-feature matrix, 25 pairs exceed $|r| > 0.85$.

PEARSON CORRELATION COEFFICIENT

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}} \equiv \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

-1 ——— 0 ——— +1
PERFECT INVERSE · NO LINEAR · PERFECT DIRECT

$|r| > 0.85 \Rightarrow$ two features carry essentially the same information



OUTSIDE THIS CLUSTER

1.0000

relative_growth_0_5h
=
area_growth_rel_0_5h

TWO FEATURES. ONE SIGNAL.

Why these five are dropped.

Each row: the dropped feature, its near-perfect twin (or "near-constant" if it had none).

The model needs one feature per redundant pair, not both. 34 → 29.

DROPPED		REASON	KEPT TWIN	
01	closing_speed_m_per_h	$r = -0.998$ REDUNDANT	→	dist_change_ci_0_5h
02	closing_speed_abs_m_per_h	$r = +0.907$ REDUNDANT	→	area_growth_abs_0_5h
03	dist_std_ci_0_5h	$r = -0.943$ REDUNDANT	→	dist_slope_ci_0_5h
04	projected_advance_m	$r = +0.855$ REDUNDANT + SPARSE	→	area_growth_abs_0_5h
05	dist_fit_r2_0_5h	near-constant NEAR-CONSTANT	—	—
PER-HORIZON BRIER IMPACT · base_full → after manual cut				
12h		24h	48h	72h
-0.034 HELPED		+0.014 HURT	-0.001 HELPED	-0.001 HELPED

5 features removed · 4 had a near-perfect twin · 1 was nearly constant

34 → 29

Then build what raw features can't see.

The raw feature `dist_min_ci_0_5h` is the minimum fire-to-target distance observed in the first 5 hours — by far the strongest single signal in the data. Two examples below show how transforming it lifts the model's importance score even further. **29** → **36**.



Which feature set should we ship?

Three candidate sets — raw 34, after the manual cut (29), after engineering (36) — scored on CatBoost validation Brier (lower is better). The engineered set wins decisively on 48h and 72h, where Brier drops 4–5× vs. the raw set.

	HORIZON 12h	HORIZON 24h	HORIZON 48h	HORIZON 72h
34 · RAW ALL FEATURES	0.123	0.047 BEST	0.010	0.005
29 · + MANUAL CUT -5 REDUNDANT (SLIDE 5)	0.089 BEST	0.061	0.009	0.004
36 · + ENGINEERED +7 TRANSFORMS (SLIDE 6)	0.097	0.059	0.002 BEST	0.001 BEST

CatBoost validation Brier - lower is better

→ SHIP THE 36-FEATURE SET

THREE RULES

- 01 Cut what's *redundant*.
- 02 Build what's *missing*.
- 03 Trust *validation*.

BETTER FEATURES. BETTER MODEL.

34 → 29 → 36 · AUC 0.9987 at 48h